

Stage 3D Physics Note: Nontrivial Hidden Metric and Ricci Ingredients

GL-with-AI Project

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1 Scope

This note summarizes the Stage 3D symbolic derivation scaffold. It is a derivation artifact only. It does not implement CCZ4 source terms, add `hww/Aww` variables, or make the current `BlackStringToy` scaffold physically meaningful.

2 Metric Ansatz

The first nontrivial hidden-metric check uses the diagonal SO(3)-symmetric spatial metric

$$dl^2 = a(x, z) dx^2 + c(x, z) dz^2 + x^2 q(x, z) d\theta^2 + x^2 q(x, z) \sin^2 \theta d\phi^2. \quad (1)$$

Here $q(x, z) = \gamma_{ww}$ is the physical hidden cartoon component. The factors x^2 and $x^2 \sin^2 \theta$ are coordinate reconstruction factors, not part of the hidden Cartesian-like component itself.

3 Christoffel Checks

The key hidden-sector Christoffels are

$$\Gamma_{\theta\theta}^x = -\frac{1}{2a} \partial_x (x^2 q), \quad \Gamma_{\theta\theta}^z = -\frac{1}{2c} \partial_z (x^2 q), \quad (2)$$

$$\Gamma_{x\theta}^\theta = \frac{1}{2} \partial_x \log(x^2 q) = \frac{1}{x} + \frac{1}{2} \partial_x \log q, \quad \Gamma_{z\theta}^\theta = \frac{1}{2} \partial_z \log q. \quad (3)$$

The ϕ versions satisfy

$$\Gamma_{\phi\phi}^x = \Gamma_{\theta\theta}^x \sin^2 \theta, \quad \Gamma_{\phi\phi}^z = \Gamma_{\theta\theta}^z \sin^2 \theta, \quad (4)$$

$$\Gamma_{x\phi}^\phi = \Gamma_{x\theta}^\theta, \quad \Gamma_{z\phi}^\phi = \Gamma_{z\theta}^\theta. \quad (5)$$

4 Flat Limit and Multiplicity

Setting $a = c = q = 1$ recovers the Stage 3C flat spherical Christoffels. For any angular tensor satisfying

$$T_{\phi\phi} = T_{\theta\theta} \sin^2 \theta, \quad (6)$$

the angular contractions agree:

$$g^{\phi\phi} T_{\phi\phi} = g^{\theta\theta} T_{\theta\theta}. \quad (7)$$

For the black-string reduction,

$$N_{\text{hidden}} = \text{GR_SPACEDIM} - \text{CH_SPACEDIM} = 4 - 2 = 2, \quad (8)$$

so these two equivalent angular sectors give two hidden contributions.

5 Ricci Structural Checks

The Stage 3D script checks the structural Ricci identities

$$\frac{R_{\phi\phi}}{\sin^2 \theta} - R_{\theta\theta} = 0 \quad (9)$$

and

$$g^{\theta\theta} R_{\theta\theta} + g^{\phi\phi} R_{\phi\phi} = 2g^{\theta\theta} R_{\theta\theta}. \quad (10)$$

It also verifies that R_{xx} , R_{zz} , $R_{\theta\theta}$, and $R_{\phi\phi}$ vanish in the flat limit.

6 Constant- q Ricci Regression

Because the general abstract Ricci expressions are large, the script also asserts a nontrivial constant- q regression. For

$$dl^2 = dx^2 + dz^2 + q_0 x^2 d\Omega_2^2, \quad (11)$$

the metric is flat only when $q_0 = 1$. The asserted Ricci components are

$$R_{xx} = 0, \quad R_{zz} = 0, \quad R_{xz} = 0, \quad (12)$$

$$R_{\theta\theta} = 1 - q_0, \quad R_{\phi\phi} = (1 - q_0) \sin^2 \theta, \quad (13)$$

and the Ricci scalar is

$$R = \frac{2(1 - q_0)}{q_0 x^2}. \quad (14)$$

This is a self-guarding Ricci check, not a completed CCZ4 RHS derivation.

The constant- q_0 regression is a useful non-flat check, but it does not test the derivative sector of the general $q(x, z)$ expressions. Since $\partial_x q = \partial_z q = 0$ and all second derivatives vanish for $q = q_0$, the terms involving gradients and Hessians of γ_{ww} remain unguarded by an internal assertion. A later derivation stage should introduce a non-constant profile $q(x, z)$ with independently known curvature, or an equivalent reference calculation, to regression-test these terms.

7 Remaining Work

The off-diagonal reduced component g_{xz} , non-constant $q(x, z)$ derivative-term regression, conformal variables χ, h_{ww} , full CCZ4 RHS terms, regularized small- x limits, C++ translation, and unit-test fixtures remain future work.